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RUBIX: Fast and differentiable modeling of IFU data

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Software

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Summary

Integral field unit (IFU) spectroscopy provides spatially resolved spectral data that are central to modern studies of galaxy formation and evolution. Interpreting such data increasingly relies on forward modeling, in which theoretical models or simulations are translated into the observational domain. However, existing IFU forward-modeling tools are typically computationally expensive, CPU-bound, and non-differentiable, limiting their applicability to large surveys and preventing their use in modern gradient-based inference and machine-learning workflows.

`RUBIX` is an open-source Python package for fast and fully differentiable forward modeling of IFU data from particle-based galaxy models. Implemented entirely in `JAX`, `RUBIX` leverages just-in-time compilation, automatic vectorization, and native GPU/TPU support to generate realistic mock IFU data cubes in seconds. Its end-to-end differentiable architecture enables gradient-based optimization, variational inference, and seamless coupling to machine-learning models, supporting both forward and inverse modeling within a single framework. `RUBIX` is designed as a modular, functional pipeline, promoting reproducibility, extensibility, and integration into simulation-based inference workflows for IFU surveys.

Statement of need

Large IFU surveys such as CALIFA, MaNGA, SAMI, GECKOS, and current and upcoming programs with instruments like VLT/MUSE and JWST/NIRSpec are producing vast, information-rich datasets that demand scalable and flexible analysis methods. Forward modeling enables direct, apples-to-apples comparisons between theoretical models and data, but existing IFU forward-modeling tools are limited in several important ways.

First, computational performance remains a major bottleneck: widely used packages for generating mock IFU observations from simulations often require tens of minutes to hours per galaxy on CPUs, making large mock surveys or extensive parameter studies impractical. Second, these tools are generally non-differentiable, which precludes efficient gradient-based optimization and inference. As a result, there are limitations to inverse modelling which must rely on expensive sampling methods or simplified approximations, limiting the scope and precision of simulation–observation comparisons. Third, many existing codes are monolithic and difficult to extend, with hard-coded modeling assumptions that hinder reproducibility and extensibility through exploration of alternative physical models.

`RUBIX` addresses these limitations by providing a fast, GPU-accelerated, and fully differentiable IFU forward-modeling pipeline. Built on `JAX`, `RUBIX` enables end-to-end automatic differentiation through all stages of the modeling process (from particle-based inputs to science-ready IFU data cubes), allowing gradient-based parameter estimation, variational inference, and integration with modern machine-learning techniques. Its modular, functional architecture facilitates reproducible workflows and straightforward integration of alternative spectral models, dust prescriptions, and instrument effects. Together, these features make RUBIX a practical foundation for large-scale mock surveys, simulation-based inference, and machine-learning applications in IFU astronomy.

State of the field

Several existing software packages perform forward modeling of IFU observations from simulations, including SimSpin (Katherine E. Harborne et al., 2020; K. E. Harborne et al., 2023), MaNGIA (Sarmiento et al., 2023), GalCraft (Wang, 2024) or Synthesizer (Lovell et al., 2025; Roper et al., 2025). These tools enable the generation of realistic mock data but are typically CPU-bound and typically require tens of minutes to hours per target and do not support automatic differentiation to e.g. enable gradient-based optimization, limiting their use in modern inference workflows.

More general forward-modeling frameworks, such as Synthesizer, provide flexible and efficient generation of synthetic observables from theoretical galaxy models, with a strong emphasis on modularity and performance. Yet they stop short of full end-to-end differentiation for IFU-specific effects. At the same time, differentiable imaging frameworks such as the JAX-based scarlet2 code (Peter Melchior, 2025) showcase the power of JAX for optimization, but target imaging rather than combined spatial–spectral data.

The field is therefore in transition: scalable, GPU-accelerated, and differentiable IFU modeling remains rare, and community benchmarks for speed, gradient fidelity, and reproducibility are only beginning to emerge. RUBIX is designed to sit in this evolving landscape by offering GPU-accelerated, differentiable IFU forward modeling with modular components and reproducible configs, enabling gradient-based science cases that current CPU-bound or non-differentiable tools cannot easily support. Thereby RUBIX addresses the distinct challenges posed by IFU spectroscopy, including the combination of spatial and spectral information and the forward modeling of particle-based galaxy simulations. By extending differentiable modeling concepts to IFU data, RUBIX fills a methodological gap between traditional forward-modeling codes and modern machine-learning–driven inference frameworks.

Software description

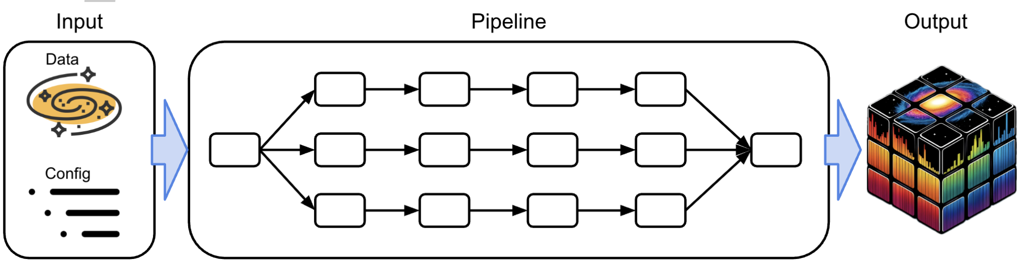

Figure 1: Schematic overview of the RUBIX software: We hand into the pipeline particle data and a configuration as input. The pipeline splits the data onto different devices. The pipeline itself consists of functiona that are applieds in a linear way. As output of our software we get an IFU cube,.

RUBIX is implemented as a modular Python package built entirely on the JAX (Bradbury et al., 2018) high-performance numerical computing framework. The software follows a functional design paradigm, expressing the forward-modeling pipeline as a composition of pure functions operating on particle-based inputs and user-defined configuration parameters. This design enables just-in-time compilation, automatic vectorization, and parallel execution across CPUs, GPUs, and TPUs without requiring user-managed low-level code.

The **forward-modeling pipeline** transforms particle data from theoretical galaxy models or cosmological hydrodynamical simulations into realistic mock IFU data cubes. Key stages include spatial binning of stellar particles, spectral synthesis based on particle age and metallicity (e.g. via FSPS (Conroy et al., 2009)), dust attenuation derived from gas distributions (with multiple dust laws available (Calzetti et al., 2000; Cardelli et al., 1989; Gordon et al., 2023)), and the application of instrumental effects such as spectral resolution, point-spread functions, and noise models. The final outputs are science-ready IFU cubes in standard astronomical formats (FITS; see Figure 1).

All stages of the pipeline are differentiable to enable **inverse modelling**. `JAX`'s automatic differentiation enables the computation of exact gradients with respect to both physical and nuisance parameters, allowing `RUBIX` to support gradient-based optimization, variational inference, and integration with modern machine-learning libraries. This key feature distinguishes `RUBIX` from traditional IFU forward-modeling tools and allows forward and inverse modeling to be performed consistently within a single framework.Comprehensive documentation, testing, and continuous integration support reproducible and extensible research workflows.

Software Design

`RUBIX` prioritizes a functional, configuration-driven architecture through the use of YAML files to keep the forward model fully differentiable while remaining extensible. The central design choice was to compose the pipeline from pure functions that accept explicit inputs (particle data and configuration objects) and return immutable outputs (intermediate tensors or IFU cubes). This reduces hidden state, simplifies gradient tracing, and makes it straightforward to swap components (e.g., alternative dust laws or SSP grids) without touching the rest of the pipeline.

We favored `JAX` transformations (jit, vmap, pmap) over custom CUDA or C++ extensions. The trade-off is an initial compile time and stricter requirements on pure, array-oriented code, but the payoff is device portability (CPU, GPU, TPU), automatic differentiation through every stage, which is essential for gradient-based inference and a massively reduced development time through high-level Python code. Where performance and clarity competed, we separated concerns: high-level orchestration lives in small, typed functions, while numerically intense steps are vectorized primitives that JIT-compile cleanly.

Instrument effects, dust models, and spectral synthesis are implemented as pluggable modules with validated defaults. This enables users to inject domain-specific choices (e.g., PSF kernels, LSF parametrizations, attenuation curves) by editing YAML configs instead of rewriting code, preserving reproducibility. The pipeline also emits intermediate products (e.g., binned particle maps, attenuated spectra) to aid debugging and validation without breaking differentiability. This design balances usability, speed, and scientific fidelity for large IFU modeling campaigns.

Research Impact Statement

`RUBIX` is already in active scientific use: it powers the generation of GECKOS mock IFU surveys (Fraser-McKelvie et al., 2024) and supports gradient-based inverse modeling in ongoing analyses. Early benchmarks on typical galaxy models show that the GPU-enabled pipeline produces science-ready cubes in about a minute, roughly an order of magnitude faster than CPU-bound alternatives such as SimSpin (Katherine E. Harborne et al., 2020; K. E. Harborne et al., 2023), MaNGIA (Sarmiento et al., 2023) or GalCraft (Wang, 2024). The codebase includes reproducible configuration files and notebooks, enabling third parties to regenerate results and adapt the pipeline to new surveys with minimal effort. Community uptake is evidenced as RUBIX has been recently presented at the Machine Learning and the Physical Sciences workshop at NeurIPS 2024 (Çakır et al., 2024) and at the 1st Workshop on Differentiable Systems and Scientific Machine Learning at EurIPS 2025 (Schaible et al., 2025). Additionally RUBIX integrates external spectral libraries (e.g., FSPS) and dust laws from the literature for easy adoption. Together these signals demonstrate both realized impact and readiness for broader adoption.

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AI usage disclosure

Generative AI tools (e.g., GitHub Copilot/ChatGPT model GPT-5.1-Codex-Max) were used to assist with aligning docstrings to implemented code, drafting portions of the documentation, refining paper wording, and suggesting code-review improvements. All other code design choices and algorithm building were solely done by humans. All AI-suggested content (code, comments, docstrings and prose) was reviewed and edited by the authors for technical correctness and appropriateness. No proprietary or unpublished data were provided to AI tools, and all scientific claims and results are derived from the authors' analyses and validated code.

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